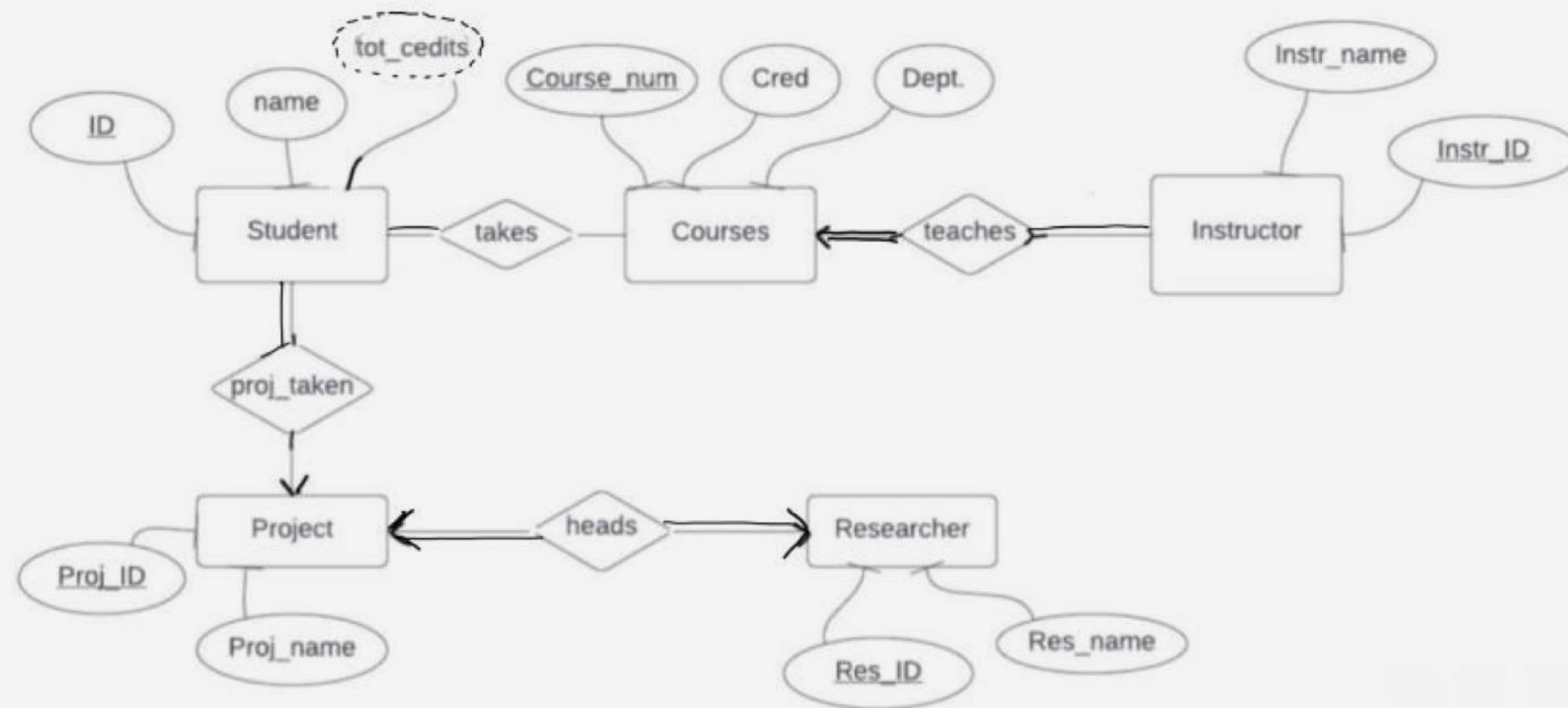


MySQL

Nested Queries | Group By | Where, Having Clause

Nested Queries



Carefully look at the cardinality mappings, to ensure that we use the correct method (amongst IN and =) while querying linked tables

Lab2 ER diagram

Nested Queries

Queries inside a query

Display the names of all students whose at least one course is taught completely/partially by "Ted Mosby".

```
mysql> select * from instructor;
+-----+-----+-----+
| instr_id | instr_name | teaches |
+-----+-----+-----+
| 1 | Sheldon Cooper | 204 |
| 2 | Phil Dunphy | 101 |
| 3 | Cameron Tucker | 301 |
| 4 | Claire Dunphy | 301 |
| 7 | Ted Mosby | 405 |
+-----+-----+-----+
```

```
mysql> select * from takes;
+-----+-----+
| stud_id | course_num |
+-----+-----+
| 1 | 101 |
| 2 | 101 |
| 5 | 101 |
| 6 | 101 |
| 7 | 101 |
| 1 | 204 |
| 2 | 204 |
| 5 | 204 |
| 1 | 301 |
| 3 | 301 |
| 4 | 301 |
| 5 | 301 |
| 1 | 405 |
| 6 | 405 |
+-----+-----+
```

```
mysql> select * from student;
+-----+-----+-----+-----+
| name | id | tot_credits | proj_id |
+-----+-----+-----+-----+
| Penny | 1 | 13 | 68 |
| Stuart Bloom | 2 | 6 | 9 |
| Howard Wolowitz | 3 | 4 | 2 |
| Will Wheaton | 4 | 4 | 5 |
| Alex D | 5 | 10 | 70 |
| Phoebe Buffay | 6 | 5 | 70 |
| Jesse Pinkman | 7 | 2 | 9 |
+-----+-----+-----+-----+
```


Nested Queries

Strategy

```
// select name of students with a particular ID
// select IDs of student who take the courses
// select courses that the instructor we want takes
```

```
select name from student where id in (
  select stud_id from takes where course_num = (
    select teaches from instructor where instr_name = "Ted Mosby"
  ));
```

In vs =

The = operator is used with Where Clause in SQL.

```
SELECT *  
FROM Student  
WHERE ADDRESS='Delhi' OR ADDRESS='Rohtak';
```

The IN operator is used with Where Clause to test if the expression matches any value in the list of values.

```
SELECT *  
FROM Student  
WHERE ADDRESS IN ('Delhi', 'Rohtak');
```

In vs =

Which ones are correct?

```
select name from student where id = (  
    select stud_id from takes where course_num = (  
        select teaches from instructor where instr_name = "Ted Mosby"  
    ));
```

```
select name from student where id = (  
    select stud_id from takes where course_num in (  
        select teaches from instructor where instr_name = "Ted Mosby"  
    ));
```

```
select name from student where id in (  
    select stud_id from takes where course_num = (  
        select teaches from instructor where instr_name = "Ted Mosby"  
    ));
```

```
select name from student where id in (  
    select stud_id from takes where course_num in (  
        select teaches from instructor where instr_name = "Ted Mosby"  
    ));
```


In vs =

```
select name from student where id = (  
  select stud_id from takes where course_num = (  
    select teaches from instructor where instr_name = "Ted Mosby"  
  ));
```

ERROR 1242 (21000): Subquery returns more than 1 row

```
select name from student where id = (  
  select stud_id from takes where course_num in (  
    select teaches from instructor where instr_name = "Ted Mosby"  
  ));
```

ERROR 1242 (21000): Subquery returns more than 1 row

```
select name from student where id in (  
  select stud_id from takes where course_num = (  
    select teaches from instructor where instr_name = "Ted Mosby"  
  ));
```

+-----+	
name	
+-----+	
Penny	
Phoebe Buffay	
+-----+	
2 rows in set (0.00 sec)	

```
select name from student where id in (  
  select stud_id from takes where course_num in (  
    select teaches from instructor where instr_name = "Ted Mosby"  
  ));
```

+-----+	
name	
+-----+	
Penny	
Phoebe Buffay	
+-----+	
2 rows in set (0.00 sec)	

Nested queries

Populate the tot_credits column in 'student'.

```
mysql> select * from courses;
+-----+-----+-----+-----+
| course_num | cred | course_name          | teaches |
+-----+-----+-----+-----+
|         101 |    2 | Intro to Break Dancing |    NULL |
|         204 |    4 | Intro to Physics for Dummies |    NULL |
|         301 |    4 | Analytical Mechanics   |    NULL |
|         405 |    3 | Intro to Real Estate   |    NULL |
+-----+-----+-----+-----+
```

```
mysql> select * from takes;
+-----+-----+
| stud_id | course_num |
+-----+-----+
|        1 |         101 |
|        2 |         101 |
|        5 |         101 |
|        6 |         101 |
|        7 |         101 |
|        1 |         204 |
|        2 |         204 |
|        5 |         204 |
|        1 |         301 |
|        3 |         301 |
|        4 |         301 |
|        5 |         301 |
|        1 |         405 |
|        6 |         405 |
+-----+-----+
```

```
mysql> select * from student;
+-----+-----+-----+-----+
| name          | id | tot_credits | proj_id |
+-----+-----+-----+-----+
| Penny         |  1 |          13 |       68 |
| Stuart Bloom  |  2 |           6 |        9 |
| Howard Wolowitz |  3 |           4 |        2 |
| Will Wheaton  |  4 |           4 |        5 |
| Alex D        |  5 |          10 |       70 |
| Phoebe Buffay |  6 |           5 |       70 |
| Jesse Pinkman |  7 |           2 |        9 |
+-----+-----+-----+-----+
```


Group By

Groups rows that have the same values into summary rows

```
mysql> select * from prof;
```

p_id	name	salary	age	course_id
1	Ramu	10000	35	3201
2	Ameya	20000	21	2109
3	Priyansh	25000	20	2210
4	Krishanu	24000	20	2210
5	Ramesh	20000	21	2210
6	Suresh	30000	22	1001
7	AJ	29000	20	2808
8	Mukesh	5000	32	1001

8 rows in set (0.00 sec)

avg(salary)	course_id
10000.0000	3201
20000.0000	2109
23000.0000	2210
17500.0000	1001
29000.0000	2808

5 rows in set (0.00 sec)

Average Salary per course

avg(salary)	age
10000.0000	35
20000.0000	21
26000.0000	20
30000.0000	22
5000.0000	32

5 rows in set (0.00 sec)

Average Salary for age groups

SYNTAX

The order is important

SELECT <column(s)>

FROM <table>

WHERE <condition(s)>

GROUP BY <column(s)>

HAVING <condition(s)>

ORDER BY <column(s)>;

What's the difference between
WHERE and HAVING?

Can we use them interchangeably?

HAVING and WHERE

- Using HAVING instead of where results in no errors
- Why Bother with WHERE?
Just use HAVING
- HAVING is slightly more computationally expensive as it does everything WHERE does + it filters groups also

```
mysql> select salary, age from prof where age > 20;
```

salary	age
10000	35
20000	21
20000	21
30000	22
5000	32

5 rows in set (0.00 sec)

```
mysql> select salary, age from prof having age > 20;
```

salary	age
10000	35
20000	21
20000	21
30000	22
5000	32

5 rows in set (0.00 sec)

HAVING and WHERE

- WHERE can't be used instead of HAVING - throws a syntax error (1064 - general syntax error)

```
mysql> select avg(salary), course_id from prof group by  
course_id having avg(salary) > 15000;
```

```
+-----+-----+  
| avg(salary) | course_id |  
+-----+-----+  
| 20000.0000 | 2109 |  
| 23000.0000 | 2210 |  
| 17500.0000 | 1001 |  
| 29000.0000 | 2808 |  
+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql> select avg(salary), course_id from prof group by  
[course_id where avg(salary) > 15000;  
ERROR 1064 (42000): You have an error in your SQL syntax  
; check the manual that corresponds to your MySQL server  
version for the right syntax to use near 'where avg(sal  
ary) > 15000' at line 1
```

Nested Queries and HAVING

Same old, same old

Similar syntax as before

- Points to keep in mind:

aggregate functions when you use GROUP BY

use HAVING when you use GROUP BY

Here - records with course_id = 2210 or 2808 will be selected.

Ramesh is selected even though his salary is not >20000 and records with course_id = 2109 aren't displayed

```
mysql> select avg(salary), course_id from prof group by course_id having avg(salary) > 15000;
```

avg(salary)	course_id
20000.0000	2109
23000.0000	2210
17500.0000	1001
29000.0000	2808

4 rows in set (0.00 sec)

```
mysql> select * from prof where course_id in (select course_id from prof group by course_id having avg(salary) > 20000);
```

p_id	name	salary	age	course_id
3	Priyansh	25000	20	2210
4	Krishanu	24000	20	2210
5	Ramesh	20000	21	2210
7	AJ	29000	20	2808

4 rows in set (0.00 sec)

Putting it all together

Select records with average salary per department > 20000 and order it by age in descending order

```
mysql> select * from prof where course_id in (select course_id from prof
group by course_id having avg(salary) > 20000) order by age desc;
+-----+-----+-----+-----+-----+
| p_id | name   | salary | age | course_id |
+-----+-----+-----+-----+-----+
| 5    | Ramesh | 20000  | 21  | 2210      |
| 3    | Priyansh | 25000 | 20  | 2210      |
| 4    | Krishanu | 24000 | 20  | 2210      |
| 7    | AJ      | 29000  | 20  | 2808      |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Group by, Having and Where + Nested queries